

Module - 2 CSS Fundamentals

CSS : Cascading Style Sheets is used to style HTML pages.

Syntax:

```
selector {  
    property : value;  
}
```

Eg:

```
h1 {  
    color: red;  
}
```

Selectors:

i) Element Selector

```
p {  
    color: blue;  
}
```

ii) Class Selector

HTML:

```
<p class = 'note'>  
Hello </p>
```

CSS:

```
.note {  
    color: green;  
}
```

one level
inside
div

iii) Child Selector

```
div > p {  
    color: blue;  
}
```

iii) ID Selector

HTML:

```
<p id = 'title'>  
Hello </p>
```

CSS:

```
#title {  
    color: black;  
}
```

iv) Universal Selector

```
* {  
    margin: 0;  
}
```

v) Descendant Selector

```
div p {  
    color: red;  
}
```

HTML:

```
<div>  
    <p> Hello </p>  
</div>
```

Specificity:
priority: Inline → ID → Class → Element

`#msg {
 color: blue;
}`

output ↗

Colors:

Red {
 i) color Name `color: red;`
 ii) color Hex code `color: #ff0000;`
 iii) color RGB `color: rgb(255, 0, 0);`

Backgrounds:

- i) Background color
- ii) Background image

`background-color: blue`
`background-image:`
`url("image.jpg");`

Borders:

`border: 2px solid black;`

thickness style color

`border-radius: 10px;` → rounded corners

Margin and Padding:

Margin → Outside space

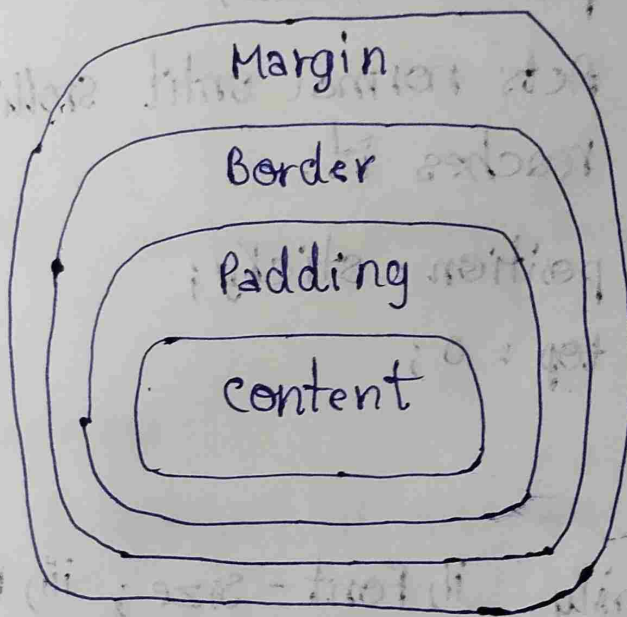
margin: 20px;

Padding → Inside space

padding: 20px;

Box Model:

Every HTML element is a box



Display Properties:

i, Block → takes full width

display: block;

ii) Inline → takes required space

display: inline;

iii) Flex → flexible layouts

display: flex;

iv) Grid → creates rows & columns.

display: grid

Positioning:

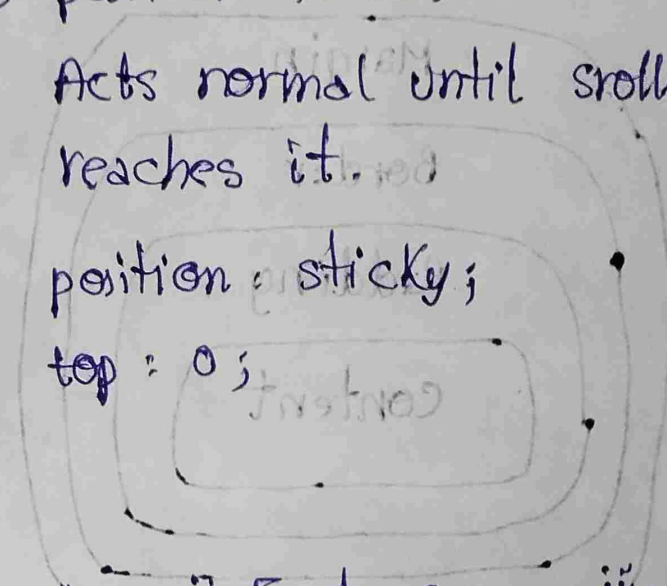
Relative: → position: relative;
moves relative to original place.

Absolute: moves relative to parent
top bottom left right
→ position: absolute;

Fixed: Stays fixed while scrolling
→ position: fixed;

Sticky: Acts normal until scrolling
reaches it.

→ position: sticky;
top: 0;



Typography:

i, Font Family — ii, Font - Size; iii, Font-weight

Google Fonts:

HTML:

<head>

<link rel="stylesheet" href="link/css?"

</head>

family = Sofia

font-family: Sofia;

Icons:

Popular libraries:

- i) Font awesome. `<i class="fa fa-home"`
- ii) Bootstrap icons. `<i class="bi bi-house">`
`</i>`

CSS variables:

- reusable values

```
:root {
```

```
--primary: #2563eb;
```

```
}
```

```
p {
```

```
background-color: var(--primary);
```

```
}
```

Transitions:

```
transition: all 0.3s ease;
```

Example:

```
p: hover {
```

```
background: red;
```

```
}
```

Shadows:

```
box-shadow: 0 4px 10px rgba(0,0,0,.2);
```

Responsive Units:

i) px

ii) %

iii) rem

↓
root-html

iv) em

↓
parent

v) vw

vi) vh

→ viewport

height

↘ Respective
of view
port

eg: font-size : 50vw

width : 50%

height : 100vh;

body {

font-size : 16px

}

h1 {

font-size : 2.5 em

→ 2.5 × 16 px

}

Flexbox Introduction:

.container {

display : flex;

justify-content : center;

~~align-items : center;~~

height : 100vh;

HTML

<div class = container>

<div> a </div>

<div> b </div>

<div> c </div>

</div>